

Using Data Mining Methods to Recognize Unusual User Activity in the Wykop.pl Community

Anomaly detection in voting and user profiles

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UNIVERSITY

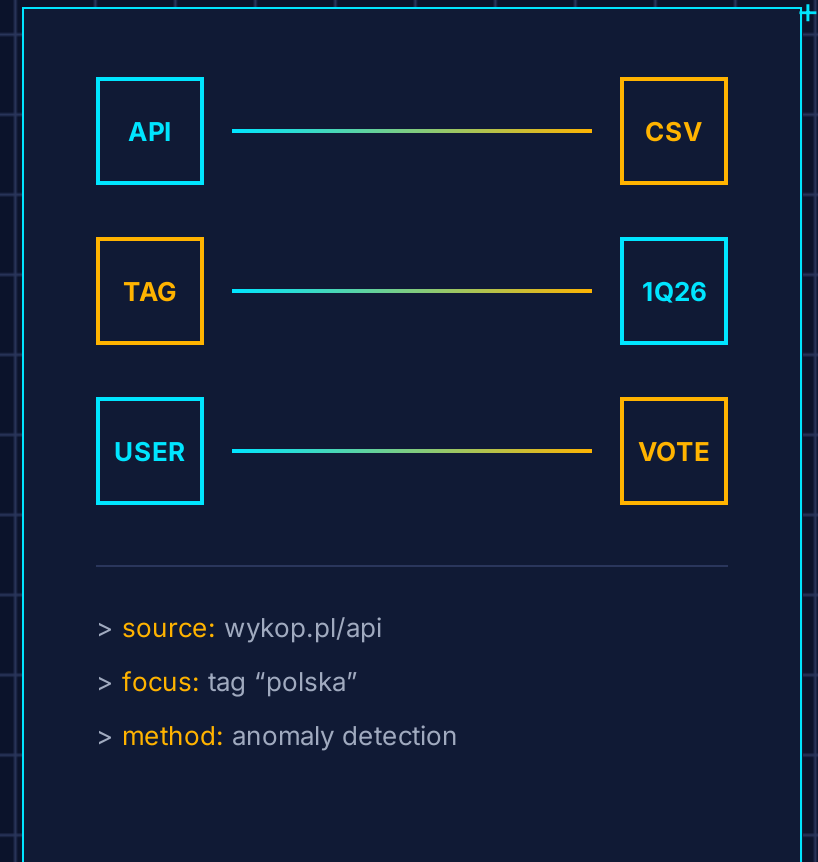
AGH University of Krakow

FIELD OF STUDY

Data Analysis – Data Science

YEAR

Krakow 2026



Purpose and context of the work

Manipulation Problem

Modern social portals are exposed to coordinated actions aimed at artificially promoting or hiding information.

Research Area

Concentration on the #polska tag, which due to its nature is particularly susceptible to attempts to steer the discussion.

Analysis Goal

Using publicly available data from the Wykop.pl API to create tools supporting anomaly detection.

EDA Approach

Application of exploratory data analysis to visualize deviations from typical user behaviors.

MOTIVATION: Ensuring the authenticity of discussion and protecting users from disinformation through technical support for moderators.

Scope and data sources

TIME PERIOD

Q1 2026

Analysis covers data from January to March 2026, capturing seasonal trends.

TOPIC

Tag #polska

One of the most popular and active discussion areas on the portal.

SOURCE

API v3

Metadata of findings and votes retrieved via official programming interface.

FIG. 1. TAG POLSKA API RESPONSE

```
tags = {}
tags_details = api.tags_get_popular_tags()
for tag in tags_details['data']:
    tags[tag['name']] = tag['followers']

print(dict(sorted(tags.items(), key=lambda item: item[1])))
```

✓ [2] 1s 730ms

```
{'pdk': 627, 'pytanie': 845, 'usa': 1227, '4konserwy': 1745, 'neuroopa': 2023,
'rozowepaski': 3075, 'f1': 8895, 'gielda': 24994, 'mecz': 29383,
'nieruchomosci': 49016, 'humorobrazkowy': 54230, 'muzyka': 71585, 'heheszki':
105012, 'polska': 128562, 'ciekawostki': 140762}
```

DATASET: Over **1,000 findings** and tens of thousands of unique user interactions.

The study relies on a real data stream obtained directly from the official Wykop.pl API, ensuring high data fidelity and research reproducibility.

Features and analytical approach

FIG. 2. POPULAR FINDINGS DATA EXTRACTION

```
tag_polska_list = []
links = set()

for month in [1,2,3]:
    page = 1
    while True:
        tag_polska = api.tags_get_stream_of_tag('polska', StreamSortType.BEST,
        page=str(page), limit=None, year=2020, month=month)

        if not tag_polska['data']:
            break

        for link in tag_polska['data']:
            if len(str(link['id'])) == 7: # znaleziiska maja 7 cyfrowe ID
                links.add(link['id'])
                tag_polska_list.append(link)

        page += 1

print(len(links))
✓ [3] 37s 851ms
1088
```

The analysis is based on multidimensional profiling of user activity to detect unnatural behavior patterns, utilizing automated data extraction scripts.

Findings and Votes

Analysis of titles, tags, and the distribution of upvotes and downvotes over time to identify coordinated spikes.

User Profiles

Study of account age, total activity, and comment history to distinguish organic users from automated agents.

EDA Approach

Exploratory approach to detecting deviations from the norm without rigid rules, allowing for discovery of new manipulation tactics.

EXPLORATION: Application of Exploratory Data Analysis (EDA) allows for the identification of suspicious accounts through visualization of extreme behaviors.

Tools and data pipeline

STAGE 01

Python + Jupyter

Iterative query building and documentation of analysis steps for reproducibility.

STAGE 02

wykop_sdk_reloaded

Handling authorization and downloading data stream from API v3.

STAGE 03

pandas

Creating tables, cleaning, aggregations, and export to CSV for analysis.

STAGE 04

Orange Data Mining

Visual exploration of distributions and anomaly detection in a GUI environment.

The process combines Python's programming flexibility with interactive visual exploration in Orange Data Mining, bridging code and visualization.

FIG. 3. API CONFIGURATION SNIPPET

```
import pandas as pd
from wykop_sdk_reloaded.v3.client import AuthClient, WykopApiClient
from wykop_sdk_reloaded.v3.types import StreamSortType, RequestType
from wykop_sdk_reloaded.exceptions import WykopApiNotFoundError

API_key = '...'
secret_key = '...'

url = "https://wykop.pl/api/v3"

auth = AuthClient()
auth.authenticate_app(API_key, secret_key)

api = WykopApiClient(auth)
✓ [1] 480ms
```

CONFIGURATION: The applied SDK enabled secure authorization and direct access to finding streams, forming the foundation of the detection process.

Data acquisition and integration

SCALE
1088 Findings

Dataset from Q1 2026 was downloaded, integrating upvotes and downvotes for the #polska tag.

PROFILING
Account Identity

Metadata for 33,424 profiles was downloaded: age, total activity, and action history.

ANOMALIES
Deleted Accounts

182 blocked or deleted accounts were identified with a specific name masking method.

FIG. 4. USER METADATA TABLE

	user	color	actions	links	links_added	links_commented	links_published	links_related	links_up	member_since	rank	gender	company	verified	banned	name	downvotes_1Q	upvotes_1Q	
0	-----	orange	595.0	109.0	0.0	0.0	0.0	0.0	109.0	2025-11-17 12:02:11	NaN	NaN	False	False	NaN	---	---	0.0	16.0
1	-----	orange	4084.0	245.0	0.0	191.0	0.0	6.0	48.0	2025-09-25 10:53:00	NaN	NaN	False	False	NaN	NaN	4.0	3.0	
2	---pollbudzaki.pl---	orange	303.0	25.0	0.0	0.0	0.0	0.0	25.0	2024-05-16 21:37:25	NaN	m	False	False	NaN	Wojciech Radziwon	0.0	1.0	
3	---	orange	62.0	27.0	0.0	0.0	0.0	0.0	27.0	2025-08-26 03:24:27	NaN	NaN	False	False	NaN	ThaIdarie	1.0	8.0	
4	---	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.0	1.0	
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	
33601	zrakzak	orange	32442.0	25322.0	0.0	0.0	0.0	0.0	25322.0	2015-09-22 22:04:51	NaN	m	False	False	NaN	NaN	0.0	304.0	
33602	zzbk	orange	17910.0	3366.0	42.0	1533.0	7.0	6.0	1785.0	2016-03-10 14:18:01	2507.0	m	False	False	NaN	NaN	0.0	16.0	
33603	zzlogin	orange	1065.0	71.0	0.0	0.0	0.0	0.0	71.0	2023-04-29 10:35:27	NaN	m	False	False	NaN	Zygi Zygot	0.0	2.0	
33604	zzxor	orange	1425.0	999.0	0.0	14.0	0.0	0.0	895.0	2021-04-01 08:10:04	NaN	m	False	False	NaN	NaN	0.0	48.0	
33605	zzzz-dogier	orange	2.0	2.0	0.0	0.0	0.0	0.0	2.0	2023-12-27 11:21:47	NaN	NaN	False	False	NaN	Zzzz Dogier	0.0	1.0	

33606 rows x 18 columns

FIG. 5. DELETED/HIDDEN ACCOUNTS

```
suspicious_accounts_deleted = ['K.....3', 'm.....o', 'M.....L', 'w.....r', 'B.....p', 'e', 'M.....m', 'F.....', 'e.....o', 'M.....s', 's.....S', 'E.....o', 'S.....r', 'c.....o', 'S.....j', 'P.....3', 'F.....y']
```

TRACE ANALYSIS: A uniform way of masking names (Fig. 5) may suggest **group account deletion** of a mass nature, characteristic of bot networks.

Key activity anomalies

FIG. 6. ALL ACTIONS PER USER (SCATTER PLOT)

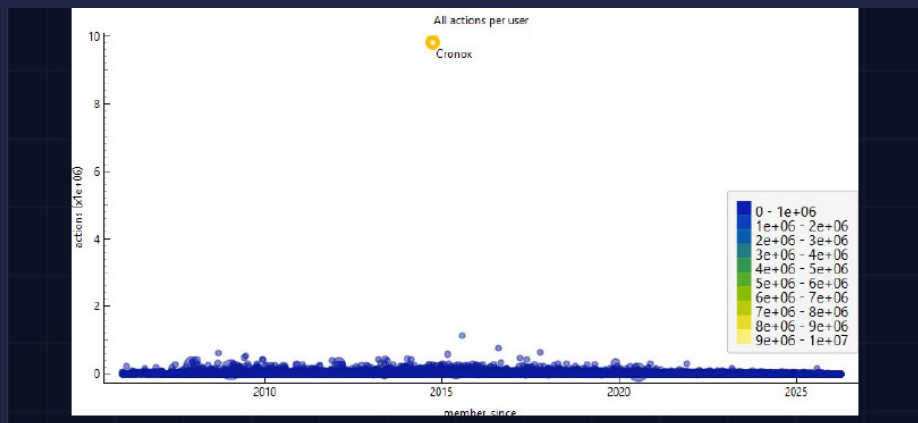
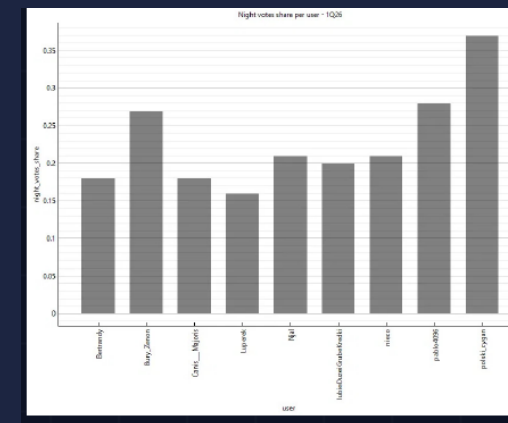


FIG. 7. TIME RHYTHM (BAR CHART)



Cronox Account

Stands out with nearly 10 million actions, suggesting a high degree of automation or technical purpose, far exceeding organic user limits.

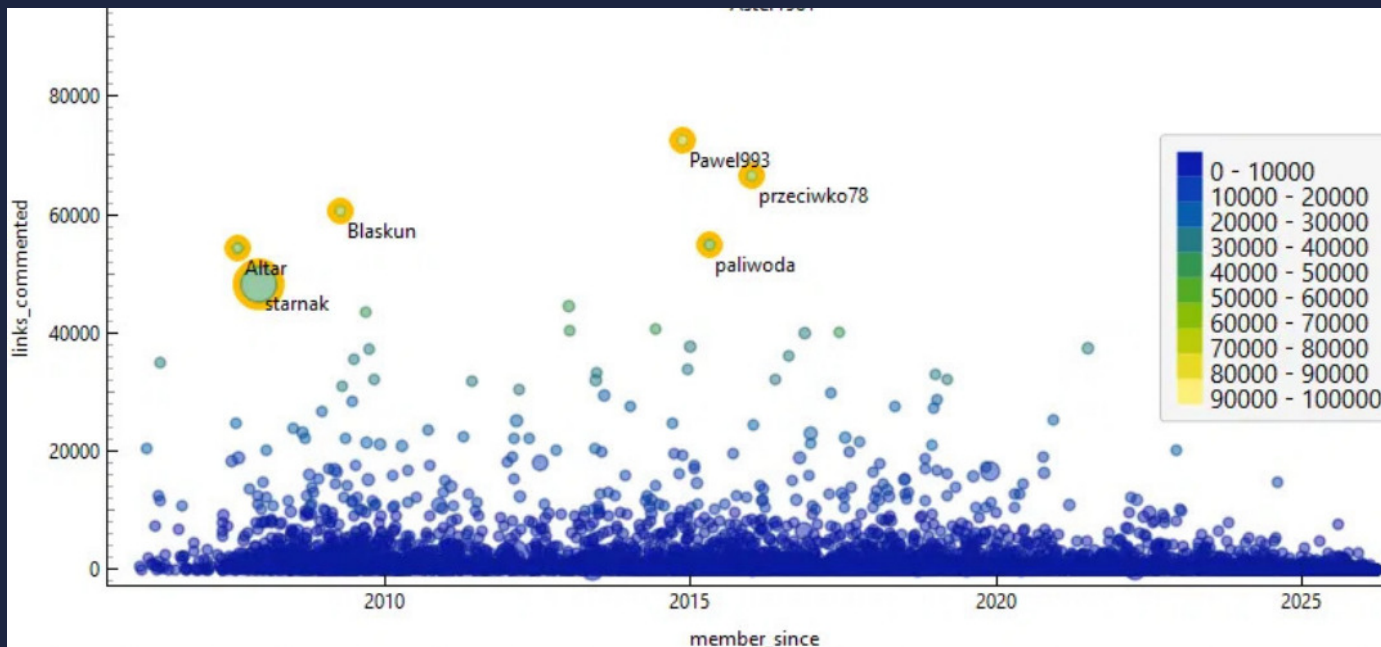
polski_cygan User

Shows the highest share of votes at night (00:00–06:00), a behavior pattern that significantly distinguishes them from typical organic users.

CONCLUSION: Extreme values on scatter plots and bar charts allow for quick identification of accounts for further, in-depth verification.

Anomalies in comments

FIG. 8. COMMENTS VS ACCOUNT AGE (SCATTER PLOT)



Correlation of the number of comments with account age reveals users with an extremely high scale of interaction, forming clear outliers in the dataset.

OUTLIER: ASTER1981

~100,000

Number of commented links, which is an absolute record in the set, indicating extreme engagement.

OUTLIER: STARNAK

~50,000

High activity with account age reaching the beginnings of the portal, acting as a historical node.

KEY CONCLUSION: Users such as Aster1981, Pawel993, or starnak act as key nodes in discussions, and their activity significantly deviates from the norm.

Extreme voting profiles

FIG. 9. UPVOTES VS DOWNVOTES (SCATTER PLOT)

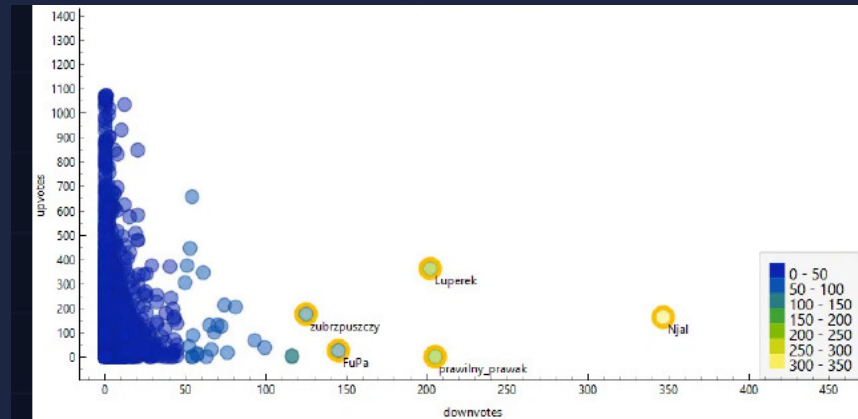


FIG. 10. EXCLUSIVELY NEGATIVE VOTES (JAN_SAH)

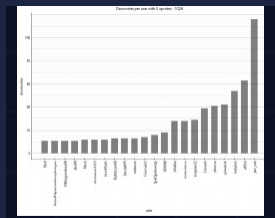
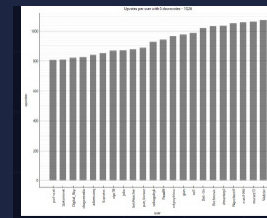


FIG. 11. EXCLUSIVELY POSITIVE VOTES (VADZIOR)



INTERPRETATION: Accounts such as **jan_sah** (116 downvotes, 0 upvotes) and **Vadzior** (over 800 upvotes, 0 downvotes) represent extreme poles of behavior in Q1 2026, acting as "filters" or "boosters".

Final conclusions

MAIN CONCLUSION

Public data from the Wykop.pl API is sufficient to meaningfully support the detection of unusual behaviors.

Results do not automatically decide on account automation but allow for pointing out profiles requiring in-depth verification.

- 01** A consistent pipeline for acquisition, cleaning, aggregation, and visualization of API data was built.
- 02** A small, influential group of accounts deviating in scale, direction, and time of activity was revealed.
- 03** The most important warning signals are one-sided voting, night activity, and extreme number of actions.
- 04** The approach can be extended to other tags, periods, and deeper analysis of coordination between accounts.

NEXT STEP: The greatest value comes from combining multiple features: activity scale, voting direction, time rhythm, and behavior repeatability. Only such multidimensional analysis allows for reliable identification of suspicious accounts.